Keystone infrastructure

Keystone

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Keystone infrastructure

Keystone infrastructure

This section describes the NetApp Keystone STaaS infrastructure, architecture, and management application for the NetApp and customer-operated environments.

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- Refresh products with new technology when deemed appropriate.
- Increase or decrease capacity of the products to meet service requirements.
- Modify architecture, technology, and/or products to meet service requirements.

The Keystone infrastructure includes multiple components, such as the following, among others:

- The Keystone infrastructure, including storage controllers.
- Tools to manage and operate the service such as AIOPs solution, Active IQ, and Active IQ Unified Manager.

Storage Platforms

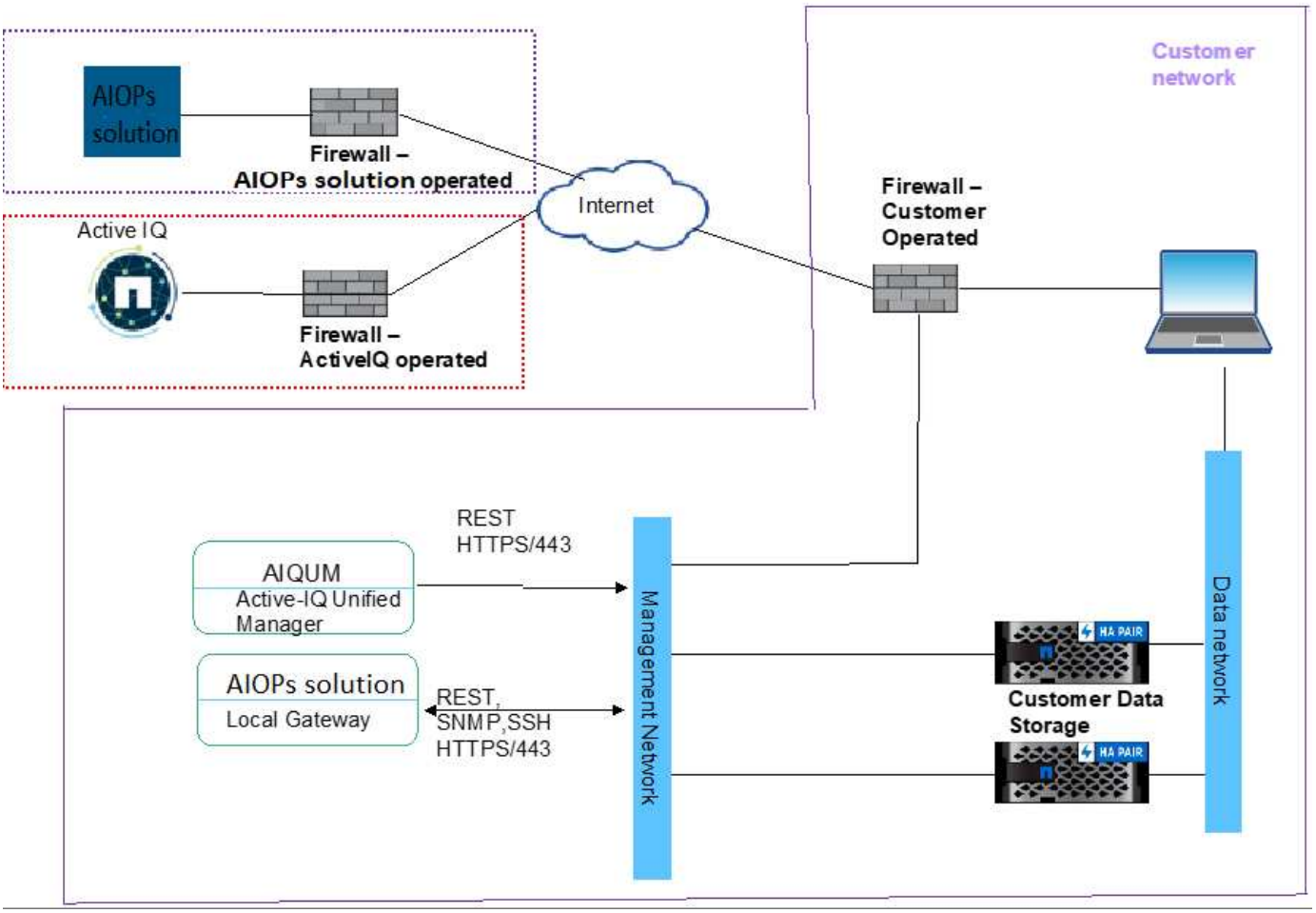
Enterprise applications need storage platforms to support fast provisioning-workflows, maintain continuous availability, sustain high workloads with low latency, deliver higher performance, and support integration with major cloud providers. NetApp has several products and technologies for supporting these requirements. For Keystone service, NetApp uses AFF and StorageGRID platforms.

Monitoring tools

In a Keystone customer-operated service, storage infrastructure and monitoring tools are installed at your site. The storage infrastructure consists of the required storage hardware needed to support your initial order, with the provision to order more storage later.

In addition to the storage equipment, two monitoring tools are provisioned for storage and consumption monitoring.

- AIOPs solution local gateway: A cloud-based application used to monitor your network. It has built-in integrations with NetApp storage platforms to collect environmental data and monitor the network. This service is enabled with the help of using a local gateway installed at your site that communicates with the cloud portal.
- Keystone Data Collector: Keystone Collector provides billing services to Keystone customers. This application is bundled with Active IQ Unified Manager. It collects data from ONTAP and StorageGRID controllers at an interval of five minutes. The data is processed, and metadata is sent to the centralized Active IQ data lake through the AutoSupport mechanism, which is used for billing data generation. Active IQ data lake processes the billing data and sends it to Zuora for billing.



Active IQ Digital Advisor (also known as Digital Advisor) enables you to log in and view the subscription and consumption details for your Keystone subscriptions. For more information about Keystone reporting on the Digital Advisor dashboard, see Keystone and Digital Advisor.

Components for deployment

This section lists the components required to enable NetApp Keystone STaaS services in your environment.

Site requirements

There are some site-specific requirements, such as space, racks, PDUs, power, and cooling, with additional network and security requirements discussed here.

Space

Floor space to host the Keystone infrastructure equipment (to be provided by customers). NetApp provides the weight specifications based on the final configuration.

Racks

Four post racks in the customer-operated offering (to be provided by customers). In the NetApp-operated offering, either NetApp or the customer can provide the racks, depending on requirements. NetApp provides 42

deep racks.

PDUs

You should provide the power distribution units (PDUs), connected to two separate, protected circuits with sufficient C13 outlets. In the customer-operated offering, in some cases, C19 outlets are required. In the NetApp-operated offering, either NetApp or the customer can provide the PDUs, depending on requirements.

Power

You should provide the required power. NetApp will provide the power requirement specifications based on 200V rating (Typical A, Max A, Typical W, Max W, Power cord type, and quantity), based on the final configuration. All components have redundant power supplies. NetApp will provide the in-cabinet power cords.

Cooling

NetApp can provide the cooling requirement specifications (Typical BTU, Max BTU), based on the final configuration and requirement.

Storage virtual machines

A storage virtual machine (storage VM) is required for the deployment of the Keystone Collector and the AIOPs solution gateway. The prerequisites for installing Keystone Collector are available here: Installation guide for Keystone Collector. The other requirements are shared during deployment.

Deployment Options

The Keystone Collector can be deployed through the following methods:

- VMware OVA template (VMware vCenter Server 6.7 or later is required)
- Customer provides Red Hat Enterprise Linux 7 or 8 or CentOS 7 Linux server. The Keystone software is installed via `.rpm` installation process.

AIOPs solution gateway is deployed on the following configuration:

- VMware OVA template (VMware vCenter Server 6.7 or later is required)
- Bootable .iso installer for
  - Citrix XenServer
  - Microsoft Hyper-V
  - Kernel-based Virtual Machine (Linux KVM)

Networking

Outbound access is required to the following services for operations and maintenance of the Keystone Collector and AIOPs solution gateway:

- support.netapp.com (usage data upload)
- keystone.netapp.com (software updates)
- Hub.Docker.io (software updates)

Depending on customer requirements and the storage controllers used, NetApp can provide 10 GB, 40 GB,

and 100 GB connectivity at the customer's site.

NetApp provides the required transceivers for NetApp-provided infrastructure devices only. You should supply transceivers required for customer devices and cabling to the NetApp-provided Keystone infrastructure devices.

Remote access requirement

Network connectivity is required between the storage infrastructure installed at the customer data center or customer owned co-located services, and Keystone operations center. The customer is responsible for providing the compute and virtual machines, and the internet services. The network design should be over a secured protocol and firewall policies will be approved by both NetApp and customers.

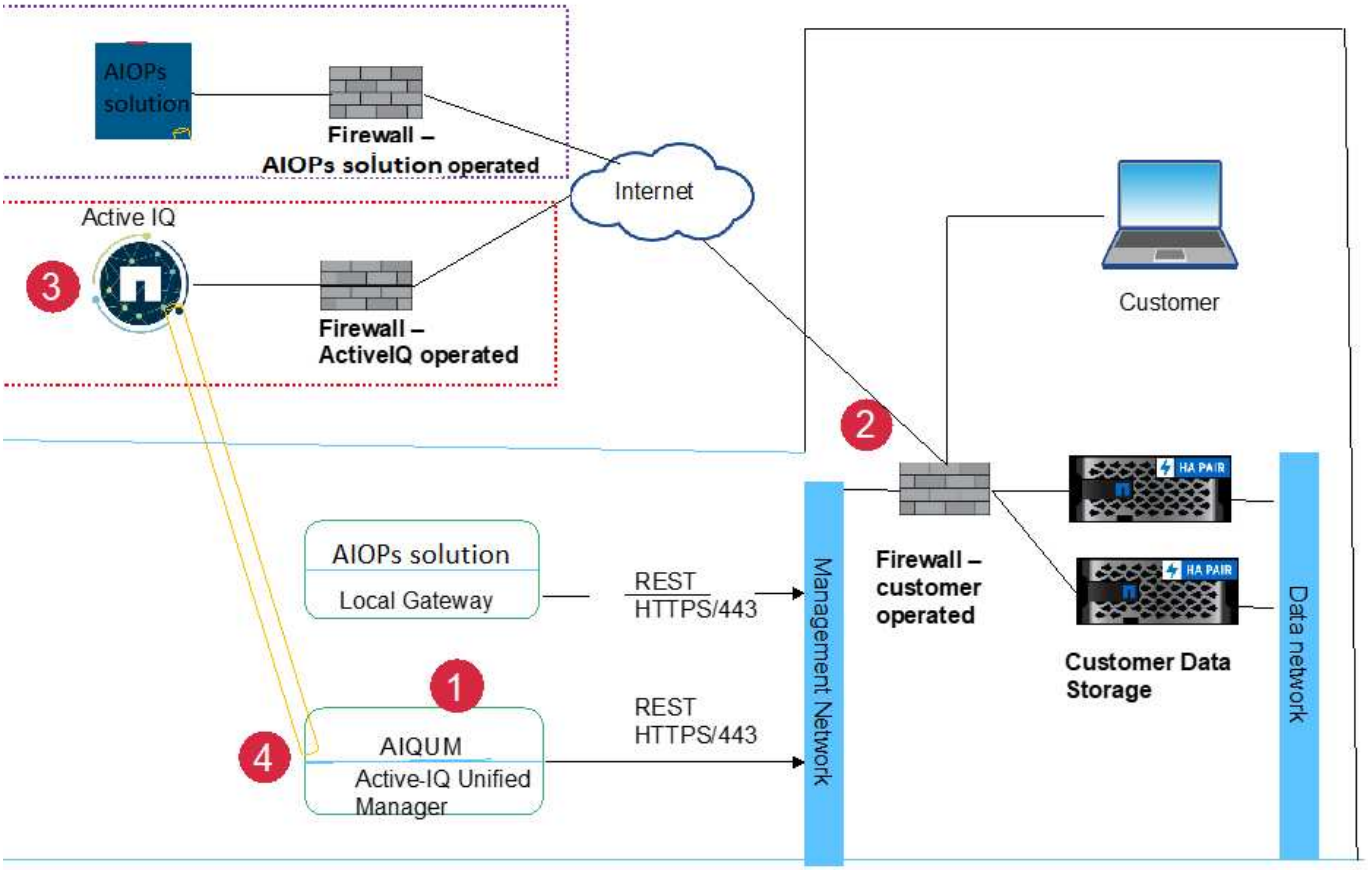
NetApp needs to access the hardware and software components installed for monitoring and management to provide services such as monitoring and billing to Keystone customers. The most common method is to establish a virtual private network (VPN) connection to the customer network and access the required data. To overcome any operational complexity perceived by customers to arise from opening firewall ports to new services, the monitoring tools initiate an external connection. NetApp cloud applications, such as AIOPs solution and Zuora, use this connection to perform their respective services. This method meets the customer requirement of not opening firewall ports though providing access to the monitoring components that are part of this service.

Keystone data flow

The data in Keystone STaaS systems flows through Keystone Collector and the associated monitoring system, the AIOPs solution tool.

Keystone Collector data flow

Keystone Collector initiates REST API calls to the storage controllers and obtains usage details of the controllers periodically, as indicated in this flow diagram:

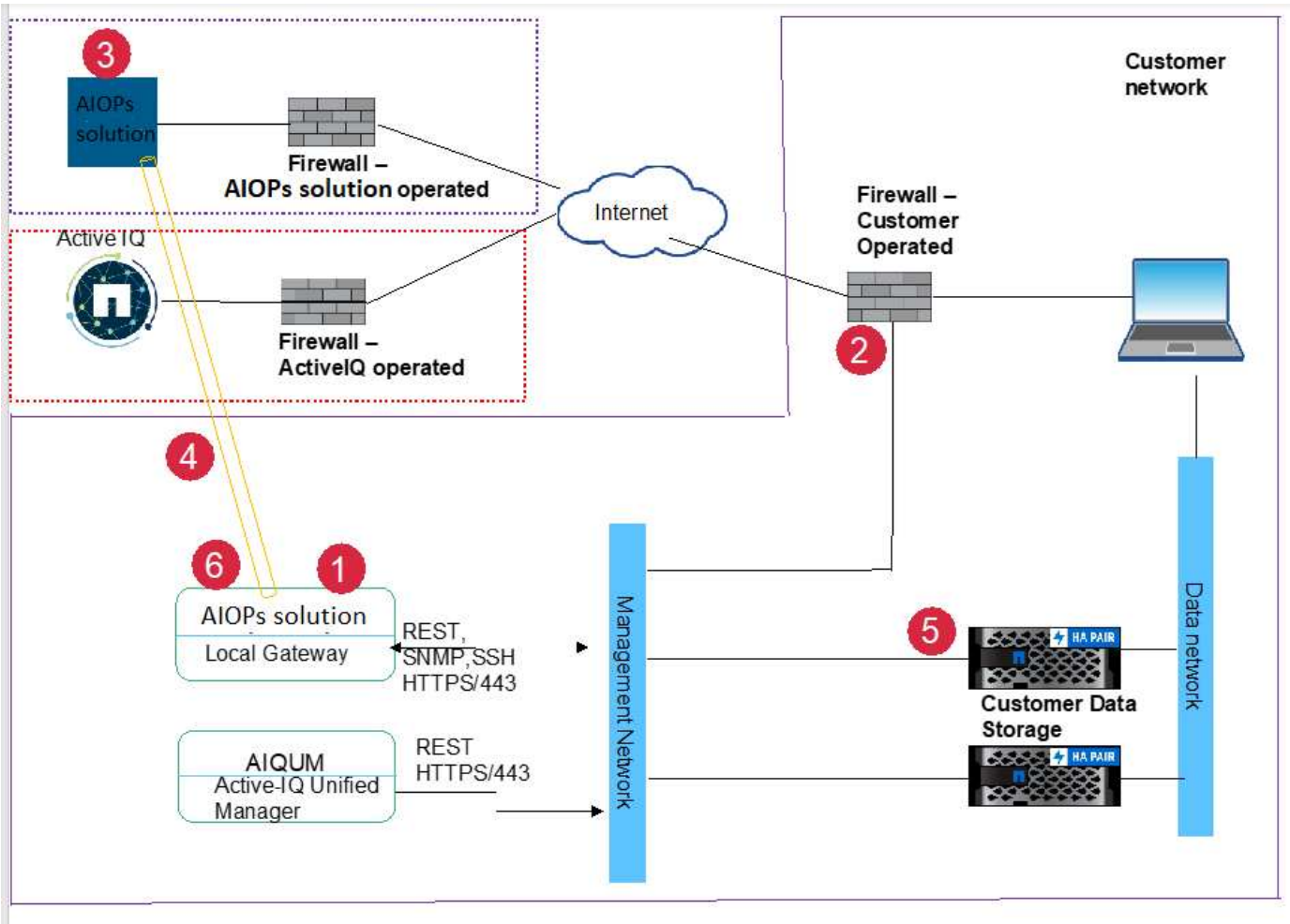


Legend

1. NetApp Collector initiates an HTTPS session to the Active-IQ cloud portal.
2. The firewall operated by the customer allows the connection.
3. The Active IQ cloud portal accepts the connection and establishes a tunnel to the NetApp Collector.
4. The NetApp collector establishes a REST API session to the management connection of the storage controller, obtains environmental data, and sends it to the Active IQ portal.

Monitoring data flows

Monitoring the health of the storage infrastructure continuously is one of the most important features of Keystone service. For monitoring and reporting, Keystone uses AIOPs solution, which needs remote access to customer's network. The following image describes how remote access to the customer location is secured by the AIOPs solution tool.



Legend

1. The AIOPs solution gateway initiates a TLS session to the cloud portal.
2. The firewall operated by the customer allows the connection.
3. The AIOPs solution server in the cloud accepts the connection.
4. A TLS tunnel is established between the cloud portal and the local gateway.
5. The NetApp controllers send alerts using SNMP protocol or respond to API requests to the local gateway.
6. The local gateway sends these alerts to its cloud portal using the TLS session, which was established before.

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